SOP-QA-017 Cleaning Validation of Product-Contact Equipment (Rev 4)

Aurora Nutraceuticals Inc. · Controlled document · For training and demonstration only

1. Purpose

This Standard Operating Procedure (SOP-QA-017) defines how cleaning procedures for product-contact manufacturing equipment are validated at Aurora Nutraceuticals Inc. Cleaning validation demonstrates, with documented evidence, that a cleaning procedure consistently reduces residues of the previous product, cleaning agents and microorganisms to predetermined acceptance limits.

2. Scope

This procedure applies to all product-contact equipment used in blending, encapsulation, tableting, coating and primary packaging at the Laval manufacturing site, including portable equipment and utensils. It does not apply to laboratory glassware, which is covered by SOP-QC-004.

3. Definitions

Dirty hold time: the period between the end of manufacturing and the start of cleaning. Clean hold time: the period between the end of cleaning and the next use of the equipment. Worst-case product: the product that is hardest to clean because of its solubility, potency or colour, selected per the product matrix in Appendix A.

4. Responsibilities

Production supervisors execute cleaning and complete the cleaning record before releasing equipment. Quality Control performs sampling and analysis. The Validation Lead writes the protocol and the report. The Quality Assurance Manager approves the protocol, the report and any deviation from this procedure.

5. Acceptance Criteria

5.1 Chemical residue: the carry-over of the previous active ingredient must not exceed 10 ppm in the next product, calculated with the maximum allowable carry-over (MACO) formula in Appendix B. Where swab sampling is used, the limit is 4 micrograms per square centimetre (4 ug/cm2) of sampled surface.

5.2 Visual inspection: all surfaces must be visually clean under normal lighting, with no visible residue, discolouration or standing water.

5.3 Swab recovery: the analytical method must demonstrate a swab recovery of at least 70 percent from the equipment material of construction before validation sampling starts. Results are corrected for the measured recovery.

5.4 Microbial limits: the bioburden on sampled surfaces must not exceed 25 colony-forming units per 25 cm2 (25 CFU/25 cm2), and no objectionable organisms may be detected.

5.5 Number of runs: a cleaning procedure is validated only after three consecutive successful cleaning runs on the worst-case product meet every criterion above.

6. Sampling

Swab samples are taken from the hardest-to-clean locations identified in the protocol, using a 25 cm2 template and the swabbing pattern in Appendix C. Rinse samples are taken from the final rinse water of closed systems. Each sample is labelled with the equipment ID, location, run number and date.

7. Hold Times

The maximum dirty hold time is 72 hours: cleaning must begin within 72 hours of the end of manufacturing. The maximum clean hold time is 7 days: equipment not used within 7 days of cleaning must be cleaned again before use. Both hold times are challenged during validation.

8. Revalidation

Cleaning validation must be repeated when the cleaning agent or its concentration changes, when the cleaning method changes, when a new product with a lower acceptance limit is introduced, after major equipment modification, and in any case every three years. A documented risk assessment may reduce the scope of a revalidation but never its acceptance criteria.

9. Records

Cleaning validation protocols, raw data and reports are quality records. They are retained according to the Records Retention and Data Integrity Policy (POL-DOC-004).